Note

While solving congruent equations, we get infinitely many solutions compared to finite number of solutions in solving a polynomial equation in Algebra.

Example 2.14 Solve $8x \equiv 1 \pmod{11}$

Solution $8x \equiv 1 \pmod{11}$ can be written as $8x - 1 = 11k$, for some integer k.

$$x = \frac{11k + 1}{8}$$

When we put $k = 5, 13, 21, 29,\ldots$ then $11k+1$ is divisible by 8.

$$x = \frac{11 \times 5 + 1}{8} = 7$$

$$x = \frac{11 \times 13 + 1}{8} = 18$$

$\therefore$ The solutions are $7, 18, 29, 40, \ldots$

Example 2.15 Compute x, such that $10^4 \equiv x \pmod{19}$

Solution

$10^2 = 100 \equiv 5 \pmod{19}$

$10^4 = (10^2)^2 \equiv 5^2 \pmod{19}$

$10^4 \equiv 25 \pmod{19}$

$10^4 \equiv 6 \pmod{19}$ $\quad (\because 25 \equiv 6 \pmod{19})$

$\therefore x = 6.$

Example 2.16 Find the number of integer solutions of $3x \equiv 1 \pmod{15}$.

Solution $3x \equiv 1 \pmod{15}$ can be written as

$3x - 1 = 15k$ for some integer k

$3x = 15k + 1$

$$x = \frac{15k + 1}{3}$$

$$x = 5k + \frac{1}{3}$$

$\because 5k$ is an integer, $5k + \frac{1}{3}$ cannot be an integer.

So there is no integer solution.

Example 2.17 A man starts his journey from Chennai to Delhi by train. He starts at 22.30 hours on Wednesday. If it takes 32 hours of travelling time and assuming that the train is not late, when will he reach Delhi?

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